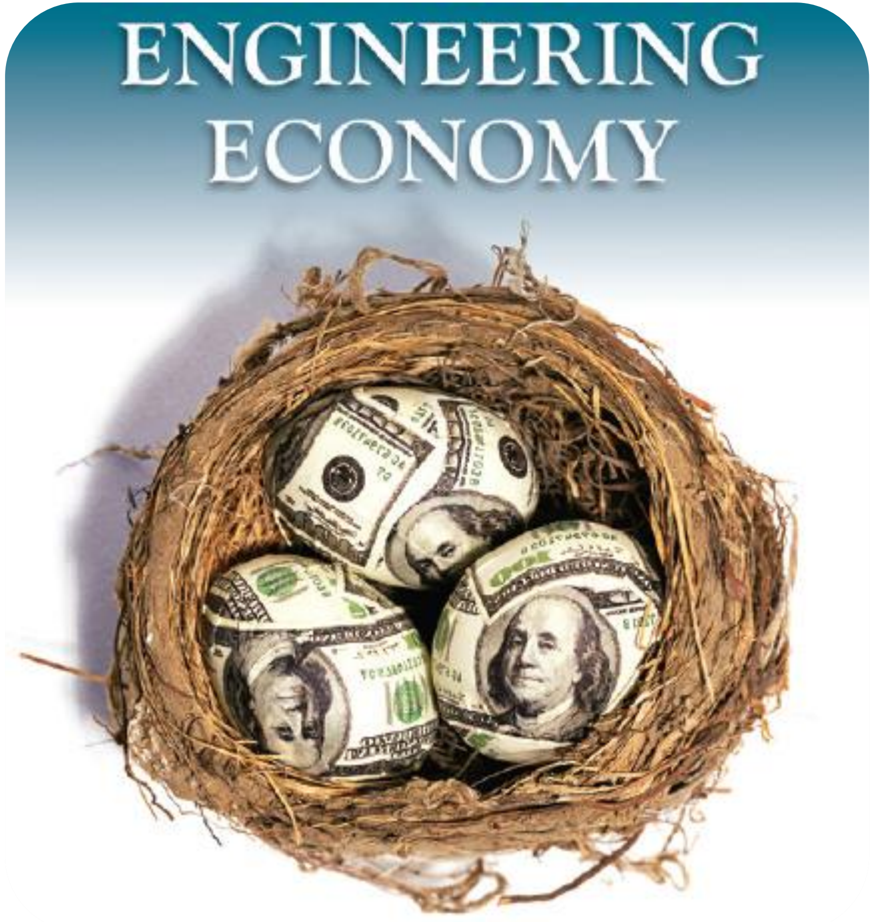
PRE 380 Engineering Economy



Chapter 2:

Factors, Effect of Time & Interest on Money

Dr. Suriyaphong Nilsang

This material is modified and based on the presentation by Blank and Tarquin (2012) 7th Edition

PRE 380 Engineering Economy

LEARNING OUTCOMES

1. F/P and P/F Factors
2. P/A and A/P Factors
3. F/A and A/F Factors
4. Factor Values
5. Arithmetic Gradient
6. Geometric Gradient
7. Find i or n

Commonly used Symbols

t = Time, usually in periods such as years or months

P = Present, value or amount of money at a time t designated as present or time 0

F = Future, value or amount of money at some future time, such as at $t = n$ periods in the future

A = Annual, series of consecutive, equal, end-of-period amounts of money

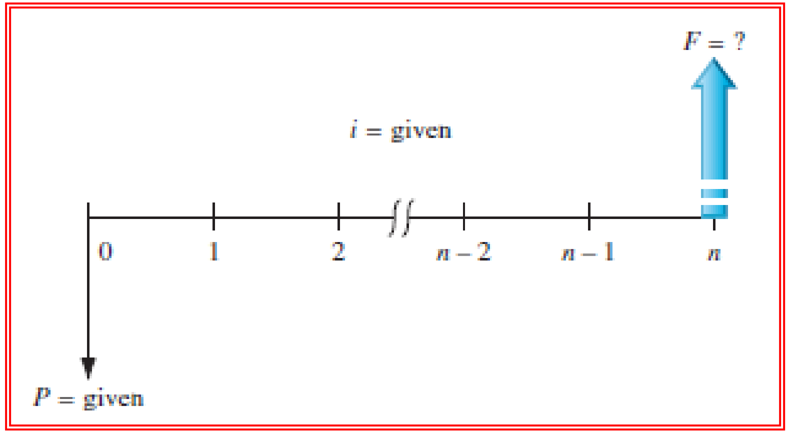
n = number of interest periods; years, months

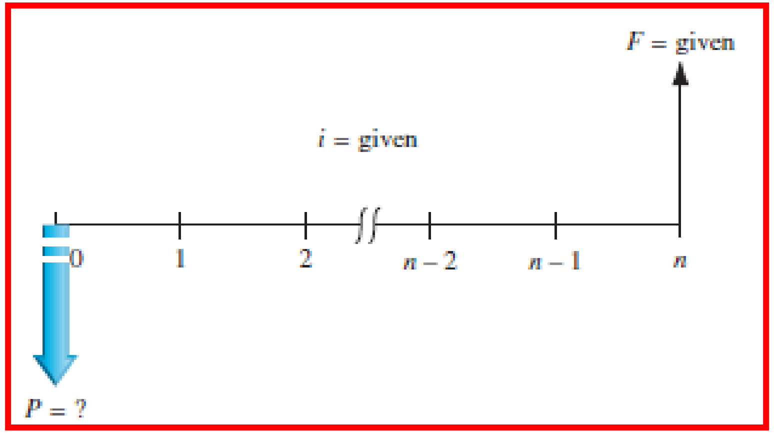
i = interest rate or rate of return per time period; percent per year or month

Single Payment Factors (F/P and P/F)

Single payment factors involve only P and F.

Cash flow diagrams are as follows:





Formulas are as follows:

$$F = P(1 + i)^n$$

$$P = F[1 / (1 + i)^n]$$

Terms in parentheses or brackets are called *factors.* Values are in tables for i and n values

Factors are represented in *standard factor notation* *such as (F/P,i,n),*

where letter to left of slash is what is sought; letter to right represents what is given

F/P and P/F Interest Factors

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8%	Compound Interest Factors							8%
	Single Payment		Uniform Payment Series				Arithmetic Gradient	
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0	0	1
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481	0.857	2
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949	2.445	3
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404	4.650	4
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846	7.372	5
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276	10.523	6
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694	14.024	7
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099	17.806	8
9	1.999	.5002	.0801	.1601	12.488	6.247	3.491	21.808	9
10	2.159	.4632	.0690	.1490	14.487	6.710	3.871	25.977	10
11	2.332	.4289	.0601	.1401	16.645	7.139	4.240	30.266	11
12	2.518	.3971	.0527	.1327	18.977	7.536	4.596	34.634	12
13	2.720	.3677	.0465	.1265	21.495	7.904	4.940	39.046	13
14	2.937	.3405	.0413	.1213	24.215	8.244	5.273	43.472	14
15	3.172	.3152	.0368	.1168	27.152	8.559	5.594	47.886	15

F/P and P/F for Spreadsheets

Future value F is calculated using FV function:

= FV(i%,n,,P)

Present value P is calculated using PV function:

= PV(i%,n,,F)

Note the use of double commas in each function

Introduction to Spreadsheet Functions

Excel financial functions

Present Value, P: = PV(i%,n,A,F)

Future Value, F: = FV(i%,n,A,P)

Equal, periodic value, A: = PMT(i%,n,P,F)

Number of periods, n: = NPER((i%,A,P,F)

Compound interest rate, i: = RATE(n,A,P,F)

Compound interest rate, i: = IRR(first_cell:last_cell)

Present value, any series, P: = NPV(i%,second_cell:last_cell) + first_cell

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Example: Estimates are P = $5000 n = 5 years i = 5% per year

Find A in $ per year

Function and display: = PMT(5%, 5, 5000) displays A = $1154.87

Example: Finding Future Value

A person deposits $5000 into an account which pays interest at a rate of 8% per year. The amount in the account after 10 years is closest to:

(A) $2,792 (B) $9,000 (C) $10,795 (D) $12,165

Example: Finding Present Value

A small company wants to make a single deposit now so it will have enough money to purchase a backhoe costing $50,000 five years from now. If the account will earn interest of 10% per year, the amount that must be deposited now is nearest to:

(A) $10,000 (B) $ 31,050 (C) $ 33,250 (D) $319,160

Example: Finding Present Value

Show present worth at present time, t = 0. If Mr.Sompong deposit $5000 at 10 years ago, $10,000 at 5 years ago, and $20,000 a year ago, which bank pays interest at a rate of 8% per year

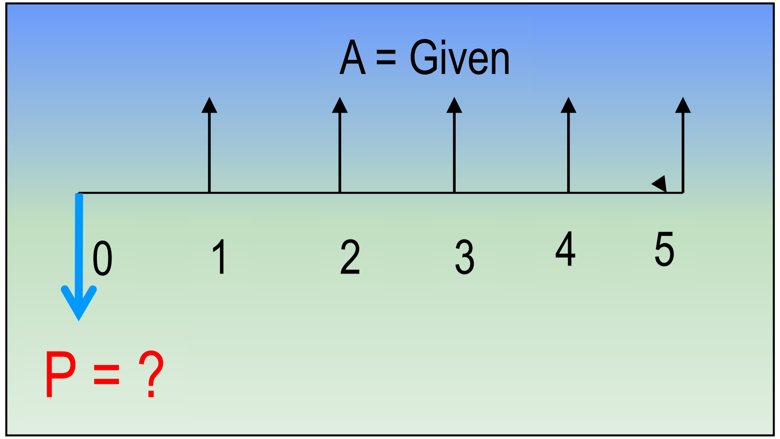
Uniform Series Involving P/A and A/P

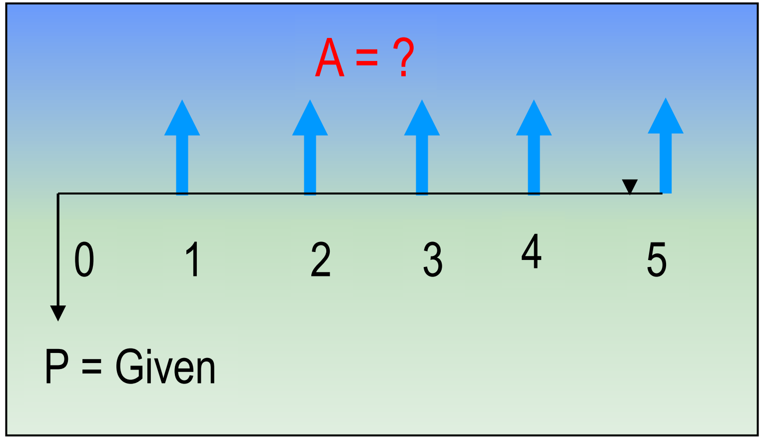
The uniform series factors that involve **P and A** are derived as follows:

(1) Cash flow occurs in ***consecutive*** interest periods

(2) Cash flow amount is ***same*** in each interest period

The cash flow diagrams are:





$P = A(P/A,i,n)$ ⟵ Standard Factor Notation ⟶ $A = P(A/P,i,n)$

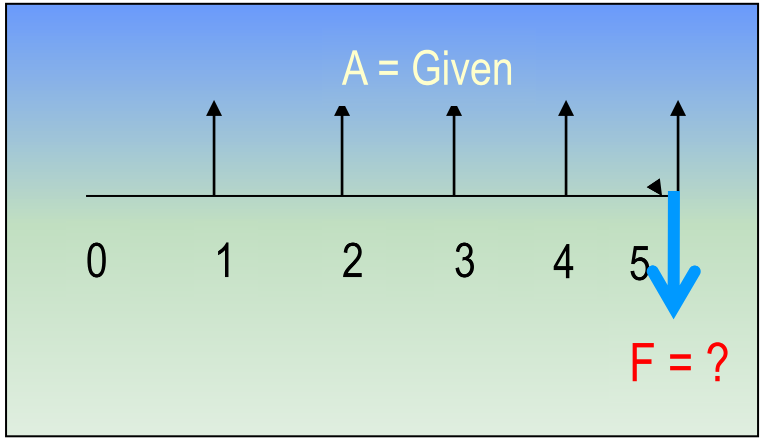
Note: P is one period *Ahead* of first A value

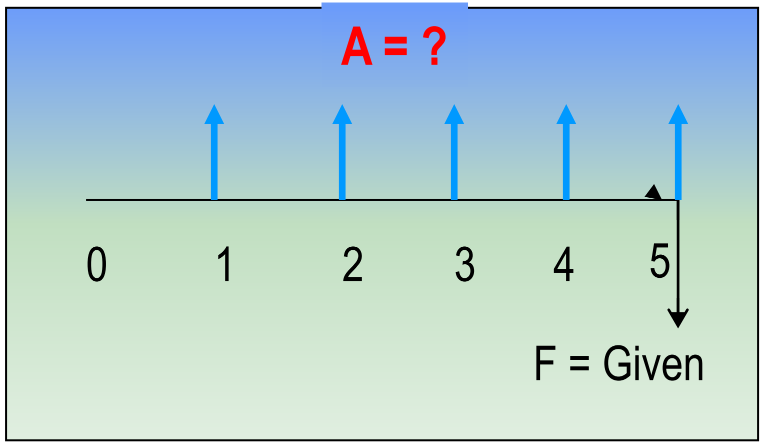
Uniform Series Involving F/A and A/F

The uniform series factors that involve **F and A** are derived as follows:

(1) Cash flow occurs in ***consecutive*** interest periods
(2) Last cash flow occurs in ***same*** period as F

Cash flow diagrams are:





F = A(F/A,i,n) ⟵ Standard Factor Notation ⟶ A = F(A/F,i,n)

Note: F takes place in the *same* period as last A

Example: Uniform Series Involving P/A

A chemical engineer believes that by modifying the structure of a certain water treatment polymer, his company would earn an extra $5000 per year. At an interest rate of 10% per year, how much could the company afford to spend now to just break even over a 5 year project period?

(A) $11,170 (B) 13,640 (C) $15,300 (D) $18,950

Example: Uniform Series Involving F/A

An industrial engineer made a modification to a chip manufacturing process that will save her company $10,000 per year. At an interest rate of 8% per year, how much will the savings amount to in 7 years?

(A) $45,300 (B) $68,500 (C) $89,228 (D) $151,500

Factor Values for Untabulated i or n

3 ways to find factor values for untabulated i or n values

- **Use formula**
- **Use spreadsheet function with corresponding P, F, or A value set to 1**
- **Linearly interpolate in interest tables**

Formula or spreadsheet function is fast and accurate
Interpolation is only approximate

Example: Untabulated i

Determine the value for (F/P, 8.3%,10)

Formula: $F = (1 + 0.083)^{10} = 2.2197$ ← **OK**

Spreadsheet: = FV(8.3%,10,,1) = 2.2197 ← **OK**

Interpolation:

8%	------	2.1589
8.3%	------	x
9%	------	2.3674

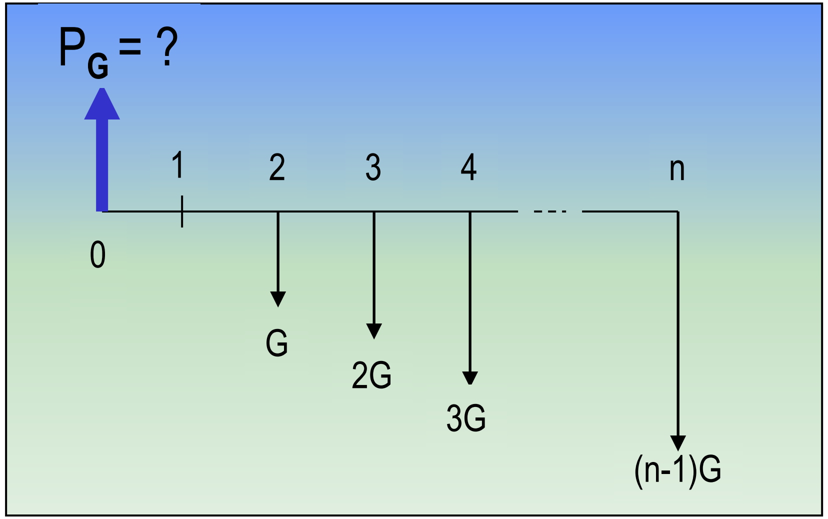
$x = 2.1589 + [(8.3 - 8.0)/(9.0 - 8.0)][2.3674 – 2.1589]$
$= 2.2215$ ← **(Too high)**

Absolute Error = 2.2215 – 2.2197 = 0.0018

Arithmetic Gradients

Arithmetic gradients *change* by the *same amount* each period

The cash flow diagram for the P_G of an arithmetic gradient is:



Standard factor notation is

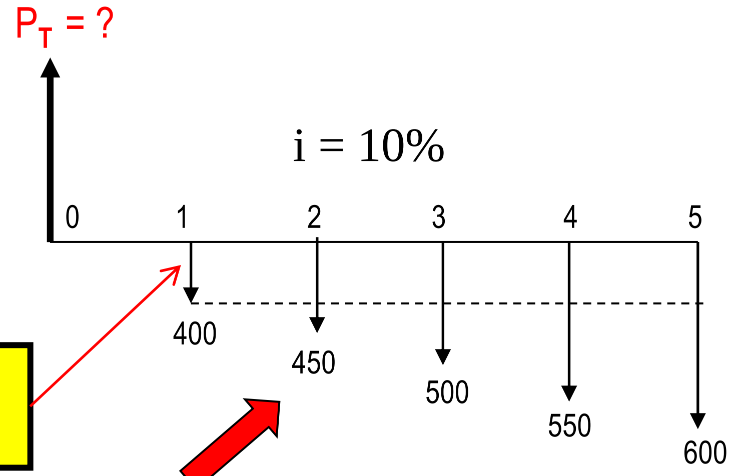
$P_G = G(P/G,i,n)$

G starts between periods 1 and 2
(not between 0 and 1)

This is because cash flow in year 1 is usually not equal to G and is handled separately as a *base amount* (shown on next slide)

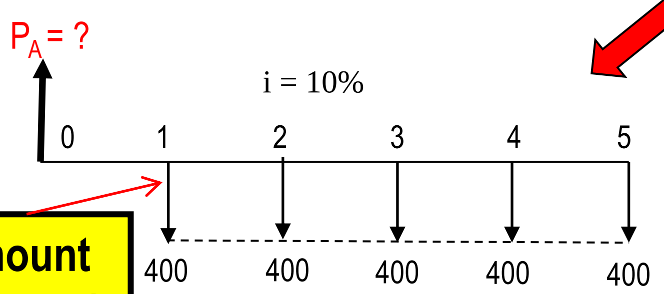
Note that P_G is located Two Periods Ahead of the first change that is equal to G

Typical Arithmetic Gradient Cash Flow



Amount in year 1 is base amount

This diagram = this *base amount* plus this gradient



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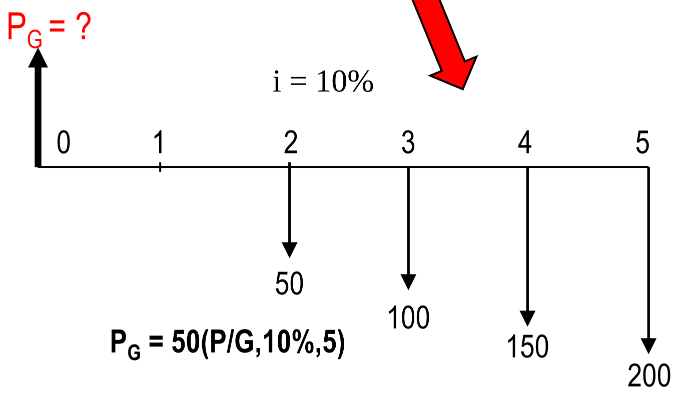


Amount in year 1 is base amount

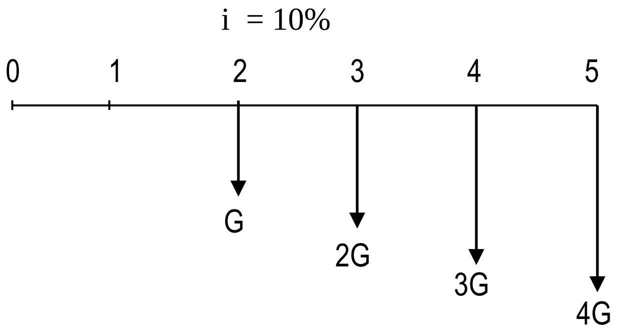
P_A = 400(P/A,10%,5)

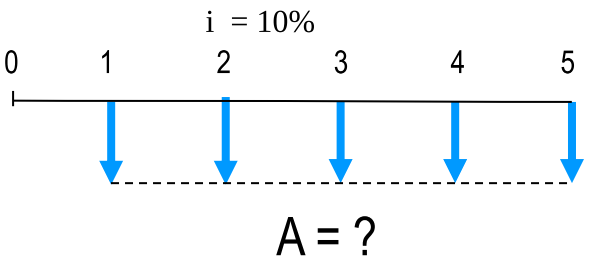
P_G = 50(P/G,10%,5)

$P_T = P_A + P_G$ = 400(P/A,10%,5) + 50(P/G,10%,5)

Converting Arithmetic Gradient to A

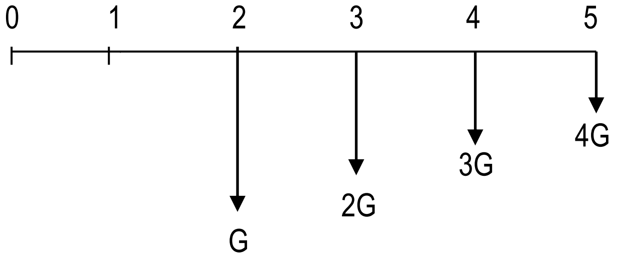
Arithmetic gradient can be converted into equivalent A value using G(A/G,i,n)





General equation when base amount is involved is

A = base amount + G(A/G,i,n)



For decreasing gradients, change plus sign to minus

A = base amount - G(A/G,i,n)

Example: Arithmetic Gradient

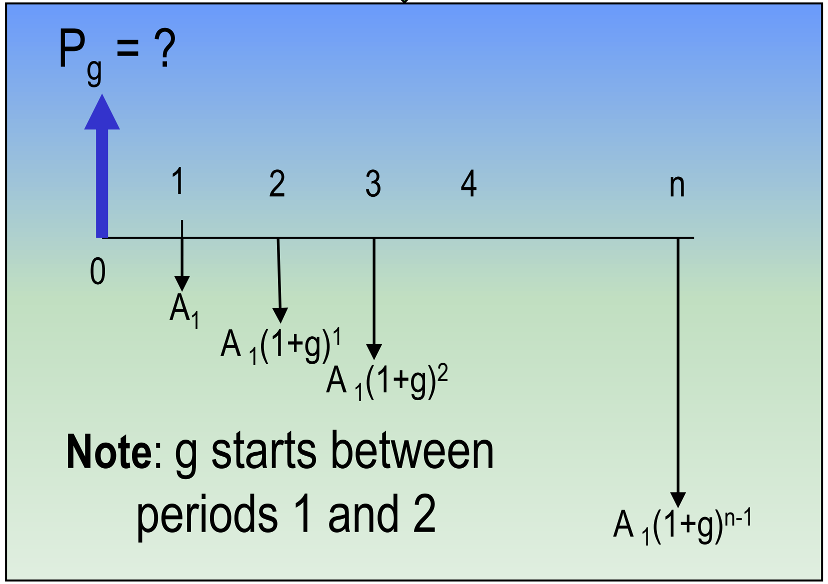
The present worth of $400 in year 1 and amounts increasing by $30 per year through year 5 at an interest rate of 12% per year is closest to:

(A) $1532 (B) $1,634 (C) $1,744 (D) $1,829

Geometric Gradients

Geometric gradients **change by the** ***same percentage*** **each period**

Cash flow diagram for present worth of geometric gradient



There are ***no tables*** for geometric factors

Use following equation for $g \neq i$:

$$\mathbf{P_g = A_1\{1- [(1+g)/(1+i)]^n\}/(i-g)}$$

where: A_1 *= cash flow in period 1*

g = rate of increase

If $g = i$, $P_g = A_1 n/(1+i)$

Note: If g is negative, change signs in front of both g values

Example: Geometric Gradient

Find the present worth of $1,000 in year 1 and amounts increasing by 7% per year through year 10. Use an interest rate of 12% per year.

(a) $5,670 (b) $7,333 (c) $12,670 (d) $13,550

Example: Geometric Gradient

One measuring instrument costs $16,000 and there is expected to be used for 6 years and a scrap value of $5,000. The first year maintenance costs are $1,700 and will increase by 11% per year. Find the total present value if given the interest rate is 8% per annum.

Unknown Interest Rate i

Unknown interest rate problems involve solving for i, given n and 2 other values (P, F, or A)

(Usually requires a trial and error solution or interpolation in interest tables)

Procedure: Set up equation with all symbols involved and solve for i

A contractor purchased equipment for $60,000 which provided income of $16,000 per year for 10 years. The annual rate of return of the investment was closest to:

(a) 15%   (b) 18%   (c) 20%   (d) 23%

Unknown Recovery Period n

Unknown recovery period problems involve solving for n, given i and 2 other values (P, F, or A)

(Like interest rate problems, they usually require a trial & error solution or interpolation in interest tables)

Procedure: Set up equation with all symbols involved and solve for n

A contractor purchased equipment for $60,000 that provided income of $8,000 per year. At an interest rate of 10% per year, the length of time required to recover the investment was closest to:

(a) 10 years (b) 12 years (c) 15 years (d) 18 years

Summary of Important Points

- In P/A and A/P factors, P is *one period ahead* of first A
- In F/A and A/F factors, F is in *same period as last* A
- To find untabulated factor values, best way is to use *formula or spreadsheet*
- For arithmetic gradients, gradient G starts between *periods 1 and 2*
- Arithmetic gradients have 2 parts, *base amount* (year 1) and *gradient amount*
- For geometric gradients, gradient g starts been *periods 1 and 2*
- In geometric gradient formula, A_1 is amount in *period 1*
- To find unknown i or n, *set up equation involving all terms* and solve for i or n